

Trading Strategies — Living Document

A running catalogue of every signal/strategy we've considered, built, tested, or shelved — with **status and evidence**, so we refine over time instead of re-litigating. Update the changelog at the bottom on every material change.

Golden rule: nothing earns risk until it survives **point-in-time features + friction-aware, regime-split backtesting**. The broad-sentiment edge looked great and died exactly this way — see `backtest/engine.py` .

Status legend

	Meaning
✔ Validated	Survived friction-aware, regime-split backtest (still preliminary — see caveats)
● Marginal	Real effect, but fragile net of costs / not sign-stable
✖ Failed	Tested and does not work (kept here so we don't retry it)
🧪 Untested	Built and wired, awaiting data to evaluate
💡 Idea	Backlog — not yet built
🔄 In progress	Data/build task underway

Progress at a glance

Last updated: 2026-06-01. Update this section whenever a status, metric, or next-step changes.

Strategy scoreboard

Strategy	Class	Status	Headline result	Next action
<code>retail_fear</code> (capitulation / bear-extreme / pessimism-fade)	Macro overlay	✅ Validated	excess vs buy-&hold , generalises across a risk-on basket: Gold +0.6 / AUDJPY +0.5 (win 88%) / SP500 +0.2pp robust; Silver +2.2pp lumpy; VIX gate ~4×	live via MT5 <code>RedditMacro_EA</code> ; vol-target sizing; longer history
<code>rrai_momentum_long</code> , <code>froth_high_long</code>	Macro overlay	❌ No edge	positive net but ~0 excess vs buy-&hold (just bull drift)	dropped
<code>rrai_euphoria_short</code> + all macro shorts	Macro overlay	❌ Failed	−1.5 to −3pp excess (shorting the bull)	dropped
<code>dump_fade_DELETION_short</code>	Equity	✅ Thesis validated (forensic) / ⚠️ not yet PIT-tradeable	gone≥0.20 → +14.3%/10d short, 83% win, t=2.57, +ve in all 3 regimes ; control proves the gate flips the trade (un-gated pump short <i>loses</i> −10.8%, t=−2.22)	make tradeable: finish author sweep to power the PIT <code>young_frac</code> gate (now n=3–7, inconclusive); express via puts
<code>fade_organic_short</code>	Equity	🟡 Marginal	+0.83%/5d but P3 flips	retry via puts + VIX/vol gate
<code>pump_long</code> (ride)	Equity	❌ Failed	+3% but regime-flips, t=0.75	— drop
<code>dump_fade_short</code> (naive)	Equity	❌ Failed	−9.9% (squeezed)	superseded by deletion-gated
<code>organic_long</code>	Equity	❌ Failed	−4.2%, sign-stable loser	inverse → <code>fade_organic_short</code>

Build & data progress

Component	Status	Detail
Feature store (<code>feature_daily</code> / <code>aggregate_daily</code>)	✅ Done	~77k ticker-day rows, point-in-time
Friction-aware backtester (<code>backtest/engine.py</code>)	✅ Done	costs + regime split + no-lookahead
<code>/signals</code> feed (multi-strategy, json mt5)	✅ Done	strategy-tagged; live, currently flat
Macro strategy R&D (<code>backtest/macro_research.py</code>)	✅ Done	only <code>retail_fear</code> beats buy-&hold; momentum/froth no edge
MT5 framework (<code>SignalClient.mqh</code> + <code>RedditMacro_EA.mq5</code>)	✅ Done	strategy-selectable thin client; demo-only
Author profiling coverage	📊 20%	30,015 / 152,040; 2,973 gone (2,087 del / 886 susp); target ~40%
Deletion-gated short testability	✅ Forensically validated	gone≥0.20 short +14.3%/10d, t=2.57 (n=30); PIT <code>young_frac</code> proxy still sparse (n=3–7)
Short-interest / squeeze data (FINRA Reg SHO)	✅ Done	<code>scraper/finra.py</code> : 475k rows / 5,693 tickers; z-score screen (abs level is MM noise, median 0.49)
Idea backlog	💡 7 ideas	squeeze · options-flow · hype-cycle · sentiment-extremes · rotation · novelty · coordination
Equity/options paper client (Alpaca)	💡 Planned	not started

Methodology (non-negotiables)

1. **No lookahead** — features at date D use only data $\leq D$; entries at the first close *after* D.
2. **Frictions** — round-trip spread (small-caps wide), short borrow, halts, capacity.
3. **Regime split** — net return per sub-period; sign-stability is the bar, not a big average.
4. **Shorts via PUTS**, not naked stock — penny shorts have squeeze/borrow/halt tails.
5. **Overlapping-window caveat** — daily signals with N-day forward returns inflate t-stats.
6. **Benchmark macro vs buy-and-hold** — in a bull market "long anything" looks like alpha; a macro long must beat *unconditional drift* (excess), not zero. (Cost us 5 false "survivors".)

Asset-class reality (what this data can/can't trade)

- **Small/micro-cap & meme equities** — primary signal (retail flow moves price).
- **Single-name options** — primary *expression* (convex longs, defined-risk shorts).
- **Index / risk-FX** — *macro overlay only* (retail positioning as a contrarian factor).
- **FX majors / "Reddit sentiment on SPY as direction"** — no edge; don't build it.

1. Macro overlay — Retail Risk-Appetite Index (RRAI)

Aggregate retail positioning as a **contrarian** factor at extremes. Built posts-only + ratios (coverage-stationary); 90-day trailing percentile. `analytics/features.py::aggregate_daily`.

Macro R&D verdict (`backtest/macro_research.py` , **excess over buy-and-hold**). A bull market made *every* macro long look like a winner and *every* short a loser — until benchmarked against unconditional drift. Net of that, only the **contrarian retail-fear** family beat buy-and-hold. RRAI-momentum and froth had ~ 0 excess (pure drift); all shorts were -1.5 to -3 pp (fighting the trend). **One real macro edge, three formulations of it.**

✓ `retail_fear` — buy risk when retail capitulates (*served live*)

- **Thesis:** retail max-bearish (RRAI pct low / bear-share extreme) + genuine fear (VIX high) $\rightarrow$ contrarian bounce. Captured 3 ways (capitulation, bear-extreme, pessimism-fade) — all positive-excess, mutually corroborating.
- **Spec (shipped):** long when `rrai_pct` ≤ 0.15 **AND** `VIX` ≥ 18 ; hold ~ 10 d; size by extremity; **fear-side only**.
- **Evidence (12mo, net, 10d): excess over buy-&-hold** SPY +0.17pp / QQQ +0.23pp / **AUDJPY +0.47pp** (win 88%); `bear_extreme` +0.24/+0.32/+0.36pp corroborates. **VIX gate $\sim 4\times$ d it** (SPY +1.56%/10d in fear vs +0.42% calm). Modest **timing tilt**, not standalone alpha (buy-&-hold already won 66–75%).
- **Instrument scan (17 MT5 instruments, 10d excess vs buy-&-hold):** the signal **generalises across the risk-on basket** — Silver +2.22pp (lumpy, one big rally $\rightarrow$ size small), Gold +0.59pp (robust, +ve every regime), AUDJPY +0.47pp (steadiest, win 88%), SP500 +0.17pp. Oil (+1.82pp but one-episode) and crypto (BTC net-negative) **dropped**. Generalisation across assets is itself evidence it's a real risk-appetite signal.
- **Expression:** a **diversified risk-on basket** — AUDJPY + Gold (XAUUSD) + US500/USTEC + small Silver (XAGUSD). Served via `/signals` (tag `retail_fear`); traded by `mq15/RedditMacro_EA.mq5` (demo).
- **Caveats:** single ~ 12 mo bull regime; **46 cap-days = only ~ 13 distinct fear episodes** (clustered $\rightarrow$ effective n small); overlapping windows; edge small in absolute terms.
- **Next:** leverage-language enrichment (calls/puts, margin/YOLO); longer history (data-limited); vol-target sizing; combine capitulation+bear-extreme into one composite trigger.

✗ Macro candidates that failed (kept so we don't retry)

- `rrai_euphoria_short` , `bull_extreme_short` , `froth_high_short` , all momentum-shorts: **-1.5 to -3 pp excess** — shorting a bull market.
- `rrai_up_momentum_long` , `froth_high_long` : **~ 0 excess** — just drift, no signal.

2. Equity cross-sectional (per ticker × day)

From `analytics/features.py::feature_daily` (velocity, accel, breadth, concentration/HHI, sentiment trajectory, author young/gone/suspended mix, novelty).

✗ `pump_long` — ride concentrated promotion

- **Thesis:** concentrated mention spike (few authors, high volume) pops short-term.
- **Result:** net +3%/5d but **sign-flips** (P1 +15.8 → P3 -4.6), win 36%, $t=0.75$. The crude "+6.45%" was a no-lookahead/no-cost/single-regime artifact. **Not robust.**

✗ `dump_fade_short` — short a decelerating pump (naive)

- **Result:** net -9.9% (P1 -29%). Shorting pumps on deceleration gets squeezed/run over. Wrong trigger; brutal without defined risk.

✗ `organic_long` — buy broad attention spikes

- **Result:** sign-stable -4.2% net ($t=-4.25$, $n=321$). Buying broad/news-driven attention tops is a *consistent loser* → the inverse is the candidate (below).

● `fade_organic_short` — fade broad attention spikes

- **Thesis:** broad attention spikes mean-revert down (inverse of the robust `organic_long` loss).
- **Result:** net +0.83%, but **flips in P3** (win 46%) — the reversion is real but shorting costs + recent weakening eat it. **Marginal.** Revisit via **puts** (cheaper than borrow) and with a VIX/vol gate.

✓ `dump_fade_DELETION_short` — short a pump whose promoters vanished (*THE thesis — forensically validated*)

- **Thesis:** concentrated pump + promoters now **deleted/suspended** → manufactured → fade the collapse. The original A2 idea, and the *reason* we track account lifecycle.
- **Result (2026-06-02, 20% author coverage, $h=10$, market-neutral, 75bps + 8bps/day borrow):** | gate | n | net short | win | t | regimes | ---|---|---|---|---|---| | `gone_frac` ≥ 0.10 | 33 | +10.0% | 81.8% | 1.51 | -1.2 / +19.9 / +11.4 | | `gone_frac` ≥ 0.20 | 30 | +14.3% | 83.3% | 2.57 | +11.4 / +27.7 / +3.9 | | **control:** pumps w/ promoters *still here* (`gone`=0) | 202 | -10.8% | 36.1% | -2.22 | -30.7 / -0.6 / -1.1 |
- The gate **flips the sign of the trade** (~+25pp spread). Un-gated concentrated pumps are *un-shortable* (they continue / squeeze); only the ones whose promoters vanish collapse. The deletion is doing the work, not pump-selection. Positive across all 3 regimes at `gone` ≥ 0.20.
- ⚠ **Not yet point-in-time tradeable.** `gone_frac` uses *hindsight*: Reddit exposes no deletion timestamp, so "gone" means "gone as of our latest sweep" — unknowable at the pump date. The PIT-available leading proxy is `young_frac` (account age at post time), but it's currently too sparse to conclude ($n=3-7$, and the few events lean the wrong way).
- **Path to tradeable:** (1) **finish the author sweep** (20%→40%+) to power the `young_frac` PIT test; (2) failing that, a **live deletion-monitor** rule — `author_status_revalidate` already detects deletions forward; short on deletion-*confirmation* within K days of a fresh pump, capturing the back half of the collapse. Express via **puts** (defined risk vs squeeze tails; many are HTB/no-borrow).

3. Idea backlog (💡 not yet built)

- **Squeeze setup (long):** Reddit squeeze chatter (Shortsqueeze/SqueezePlays) + **short pressure** → squeezable. *Partially built:* `scraper/finra.py` + `squeeze_screen()` flag tickers with an anomalous **z-score jump** in daily short-ratio vs their own baseline (HTZ/GME currently surface). *Still needs* true SI%/float/days-to-cover (FINRA **bi-monthly short-interest** report) + borrow-fee for the level; the daily file's absolute level is market-maker noise. *Not yet backtested.*

- **Options-flow signal:** parse contract-level chatter (unusual_whales/options) → retail-gamma/positioning proxy. *Needs contract parsing.*
- **Hype-cycle lifecycle:** model attention as nascent→acceleration→peak→fade; long the *organic* acceleration, fade the *concentrated* peak. Breadth-vs-concentration is the discriminator.
- **Conditioned sentiment extremes (single-name):** linear sentiment is dead, but *extreme concentrated euphoria at a price high* (short) / *capitulation after a sharp drop* (bounce) may be non-linear edges.
- **Cross-sectional attention rotation:** rank meme universe by abnormal velocity z-score; long top-decile / short bottom-decile, beta-hedged with SPY; baskets from co-mention clusters.
- **Novelty / first-mention alpha:** the "discovery" moment (silent → sudden organic burst).
- **Coordination graph (short):** rigorous version of the deletion short — cluster density + temporal bursts + duplicate text + new-account% + gone%.

4. Data gaps blocking strategies

Gap	Unblocks
Intraday prices (have EOD only)	timing pump/dump entries/exits (they move intraday)
Short interest / float / borrow fee	squeeze <i>level</i> (FINRA daily short-volume now in via <code>scraper/finra.py</code> , but it's MM noise at the level; need bi-monthly SI%/float + borrow)
Options chains / IV	options expression; options-flow signal
Longer history (data-limited to ~12mo)	true multi-regime validation of the RRAI overlay
Author coverage (20% → target ~40%+)	a <i>point-in-time tradeable</i> deletion short — thesis already forensically validated; PIT <code>young_frac</code> proxy needs more coverage

5. Execution stack

- `/signals feed` (`analytics/signals.py`, GET `/signals json|mt5`) — live, serves the capitulation overlay. ✅
- **MT5 EA** (`mql5/RedditRRAI_EA.mq5`) — thin client for the index/FX overlay; demo-only. ✅
- **Alpaca / IBKR paper client** — for the equity/options legs. 🕒 planned.

★ MULTI-REGIME VALIDATION (2020-2026, incl. COVID crash + 2022 bear)

The deep-history backfill let us test across regimes — the result that actually moves confidence: - **AUDJPY capitulation-long is REGIME-ROBUST** — positive in EVERY year 2020-2026, **including the 2022 bear** (+0.7%, win 60%), win rates 60-93%. Graduates from preliminary → **validated**. This is the core tradeable edge. - **Equity (SPY) capitulation-long FAILS in a sustained bear** (2022: -0.7%/10d, win 43%) — falling-knife problem → **must gate with the 200-DMA trend filter** (don't buy fear in a downtrend). Confirmed needed. - **Euphoria-short only works in the bear** (2022 +1.3%; loses every bull year) → shorts are **regime-conditional**, switch on only in confirmed downtrends. (Vindicates not building shorts earlier.) - Caveats: small n/year (18-60), overlapping windows, historical RRAI is a WSB-heavy sampled backfill.

Session findings — sector / calendar / latency (2026-06-01)

- **Sector segmentation rescues the pump signal** (`backtest/sectors.py`). The broad pump was a wash because it *mixed opposites*: **small-cap concentrated pumps FADE** (−14.6%/10d excess, win 22%, $t = -3.93$ — pump-and-dumps reliably reverse → a short/avoid), while **biotech pumps are a fat-tailed lottery** (+66% mean, win 48%, $t = 2.23$ — a few catalyst moonshots). Thematic universes (ai/quantum/space/nuclear/meme/frontier) had too few concentrated-pump events to judge.

- **Pump detection is TOO SLOW** (`feature_daily`). Median `novelty_days` at flag = **152**; only **1%** of pumps flagged within 3 days. Root cause: the `min_total ≥ 20` universe filter + trailing z-windows mean a ticker must accumulate ~20 mentions before it's even eligible → we catch *established-ticker spikes*, not fresh pump-and-dumps. Only **33 fresh day-0 bursts** exist in the data. **Fix:** a fresh-burst fast-path that scans raw mentions (ticker's first day with a burst), bypassing the 20-mention gate — for LIVE catching, not just post-hoc.
- **Calendar effects (10y, SPY/QQQ/Gold; absent in AUDJPY-FX):** turn-of-month is real (SPY +0.49 vs +0.20%/5d, QQQ +0.58 vs +0.35, Gold +0.38 vs +0.25); September weak (−0.2%); OpEx week underperforms. Generic equity-flow seasonality → useful as *overlays/filters*, not Reddit alpha.
- **AU sentiment track** building (`au_aggregate_daily` , post-level) to test AU-sentiment → AUDJPY.

★ **MASTER BACKTEST vs BUY-AND-HOLD** (`backtest/run_all.py` , full 2020-2026, 10d, net of costs)

Every signal judged on EXCESS over its instrument's unconditional drift (buy-and-hold-anytime). | Strategy | n | net% | B&H% | excess | win | verdict | |---|---|---|---|---|---| | retail_fear **AUDJPY** | 338 | +0.53 | +0.27 | **+0.25** | 69% | beats (modest, most robust) | | retail_fear SPY | 338 | +0.54 | +0.59 | −0.05 | 61% | **no edge** | | retail_fear Gold | 338 | +0.83 | +0.72 | +0.10 | 57% | marginal | | euphoria_short SPY | 351 | −1.26 | +0.59 | −1.84 | 27% | no edge | | smallcap_pump_fade | 87 | +9.07 | +0.59 | +8.48 | 64% | beats but single-regime + brutal costs | | biotech_catalyst | 91 | +22.2 | +0.59 | +21.6 | 36% | fat-tailed LOTTERY (low win) | | attention_fade | 252 | −0.73 | +0.59 | −1.31 | 40% | no edge |

Recalibration: with a proper buy-and-hold benchmark over the full period, the AUDJPY edge is a **modest +0.25pp/10d** (earlier raw "positive every year" overstated it - that was largely drift). SPY capitulation does NOT beat buy-and-hold. The big-excess equity signals are single-regime (smallcap) or low-win lotteries (biotech). Net: one modest-but-real macro edge, several promising- but-unvalidated screens. Do NOT size up on these as-is.

DATA WE NEED (to validate/strengthen — the binding constraints)

Data	Unblocks	Why we lack it
Intraday prices (we have EOD only)	pump entry/exit timing; fresh-pump fade execution (pumps move intraday)	not collected
Historical penny-stock prices + per-ticker features	multi-regime validation of <code>smallcap_pump_fade</code> & biotech (only 2025-26 exists)	delisted pennies → Yahoo gaps; per-ticker <code>feature_daily</code> only recent
Short interest / float / borrow fee	squeeze setups; realistic short cost/feasibility	FINRA daily short- <i>volume</i> in (<code>scraper/finra.py</code>); still need SI%/float/borrow (Ortex / FINRA bi-monthly SI)
Options chains / IV	express fades via PUTS (only sane way to short pennies); options-flow signal	not collected
More years of RRAI history	tighten the macro edge (effective n ≈ a few dozen fear episodes)	Reddit/arctic-shift depth ~2017+
Broader author coverage (~20% profiled)	a <i>PIT-tradeable</i> deletion/young-account short (thesis already validated forensically; <code>young_frac</code> proxy starved)	rate-limited Reddit /about sweep
Independent sentiment sources (StockTwits, news, Discord, crypto + AU subs)	independent episodes → real effective-sample growth without waiting calendar time	not integrated (AU track building)
Real transaction-cost data (per-instrument spreads/borrow)	trustworthy net returns (small-cap costs are modelled/optimistic)	not sourced

Update — complete-data confirmation (2026-06-01)

- US backfill finished: continuous 2020-2026 (2,344 days, 341 capitulation days). **Capitulation-long multi-regime result HOLDS on the complete series** — AUDJPY positive every year incl. 2022 bear (win 60-83%).
- `smallcap_pump_fade` is strong (+12.4% excess short, win 77%) but **only 2025-26 data exists** (n=87) — no historical penny-stock prices/features, so it **can't be multi-regime validated** → stays ADVISORY, NOT promoted to tradeable. (Single-regime; can't be fixed by backfill — penny price history doesn't exist.)

Changelog

- **2026-06-02 — Deletion-gated short forensically validated.** Author coverage 8%→20% (2,973 gone). Re-backtest: `gone_frac ≥ 0.20` → +14.3%/10d short (83% win, t=2.57, +ve all regimes); the *control* (un-gated concentrated pumps) *loses* -10.8% — so the deletion gate flips the trade, confirming the A2 thesis. **Caveat:** `gone_frac` is hindsight (no Reddit deletion timestamp) → not yet PIT-tradeable; the `young_frac` PIT proxy is still too sparse (n=3-7). Also shipped `scraper/finra.py` (FINRA Reg SHO daily short-volume, 475k rows) + z-score `squeeze_screen()` — learned the *absolute* short-ratio is market-maker noise (median 0.49), so the screen uses per-ticker z-score anomaly + chatter.
- **2026-06-01 — Instrument scan** (17 MT5 instruments). `retail_fear` generalises across the risk-on basket; commodities express it best (Silver +2.2pp lumpy, Gold +0.6pp robust), AUDJPY steadiest. Oil/crypto dropped. Basket = AUDJPY + Gold + indices + small Silver; EA + feed updated.
- **2026-06-01 — Macro R&D + MT5 framework.** Benchmarked candidates vs buy-and-hold: only the `retail_fear` contrarian family beats it (best AUDJPY +0.47pp/10d excess); momentum/froth = no edge, all shorts fail. Built `SignalClient.mqh` + `RedditMacro_EA.mq5` (strategy-selectable), multi-strategy `/signals` feed; retired `RedditRRAI_EA.mq5`.
- **2026-06-01 — Initial document.** Phase 0 (feature store + friction-aware backtester) shipped. RRAI capitulation-long validated (best via AUDJPY, VIX-gated); pump-ride/naive-dump-short/organic-long failed; fade-organic marginal; deletion-gated short awaiting author coverage (8%). `/signals` + MT5 EA live.